Capstone Project – I: Review – 2 on

Integrated Patient care dashboard for enhancing communication, medical record management, and service navigation within hospitals

Presented by the Batch No. : CSE - 39

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Feedback from Guide/Reviewers and actions taken after Review–1

Feedback from Guide/Reviewers

1. The problem statement needs to be changed as it is too confusing.
2. The objectives should be clarified, especially whether we are building a hospital management system or doctor and patient dashboards.

Actions taken

1. Simplified the problem statement to make it clearer and easier to understand.
2. Refined and clearly defined the objectives to specify the scope and boundaries of the system we are building.

Problem Statement

Integrated patient care dashboard for enhancing communication, record management, and service guidance within hospitals.

Objectives

- To provide patients with a centralized dashboard for viewing medical reports, personal documents.
- To enable doctors to efficiently access patient information, upload digital reports, and manage appointments through a unified doctor interface.
- To guide patients within the hospital by mapping departments, wards, and test locations, enabling easy navigation to required services.
- To integrate an LLM-powered assistant that analyzes all patient records and documents to generate accurate summaries, insights, and conversational explanations within the patient dashboard.

Requirement Analysis

Hardware Requirements

- Processor: Intel Core i5 or higher
- RAM: Minimum 16 GB
- Storage: 250 GB HDD / SSD – AWS S3
- GPU: NVIDIA GPU

Requirement Analysis

Software Requirements

- Operating System: Windows 10 / 11 or Linux (Ubuntu)
- Programming Language: Python 3.x
- Frameworks: FastAPI, Huggingface
- Database: MongoDB / Redis
- Frontend: React.js, Next.js
- Backend: FastAPI, n8n
- Tools: VS Code, GitHub, Postman, MongoDB Compass, MongoDB Atlas, AWS
- Browser: Google Chrome / Edge

Identification of tools/technology/algorithms

Tools

- **Git and GitHub** – For version control and CI/CD Pipelines.
- **Postman** – For API testing and backend integration verification.
- **Visual Studio Code** – For frontend and backend coding.
- **MongoDB Compass** – For database access and visualization.
- **MongoDB Atlas** – For cloud based storage.
- **AWS S3** – For storing pdfs and other files.
- **n8n –** For Agentic workflows and automatic machine learning workflow.

Identification of tools/technology/algorithms

Technologies

- **Frontend:** React.js – to build a dynamic, responsive user interface.
- **Backend:** Python with FastAPI and n8n / flowise – to handle application logic, routing, and ML model integration.
- **Database:** MongoDB, Redis – to store user summarizations, and user chats.
- **AWS S3:** For storing user reports and documents.
- **Machine Learning Environment:** Huggingface and Gemini SDK– For fine tuning the model for specific use case.
- **Deployment:** AWS / Render – for hosting the system online and ensuring scalability.

Identification of tools/technology/algorithms

Algorithms

1. Text Processing & NLP Algorithms:

•TF-IDF Vectorization – to extract important keywords from symptom descriptions.

•Sentence-BERT / MiniLM Embeddings – to convert user text into numerical vectors for better similarity matching.

•Rule-based NLP Normalization – for spelling correction, synonym matching, and multilingual term normalization.

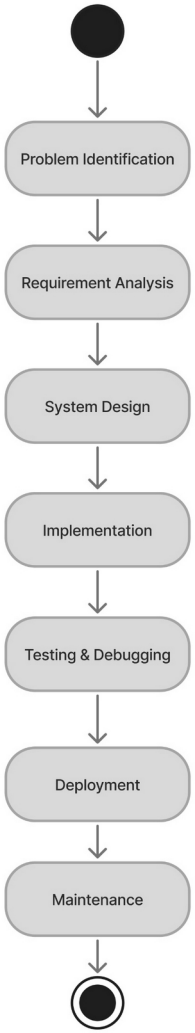
2. Machine Learning Algorithms (for Symptom → Diagnosis)

•XGBoost / LightGBM – for predicting disease categories from user symptoms

•Random Forest / Logistic Regression – baseline classifiers for tabular + text features

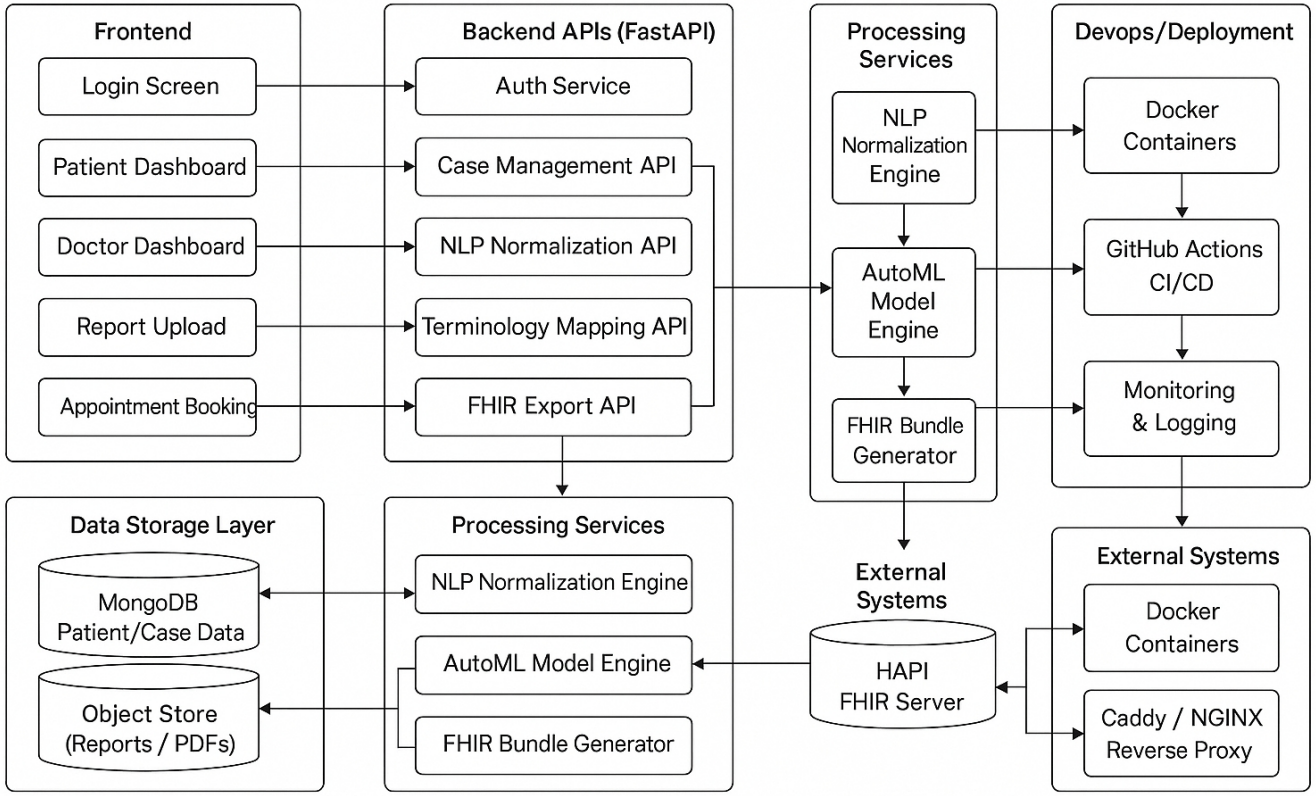
Design Strategy

Project Workflow



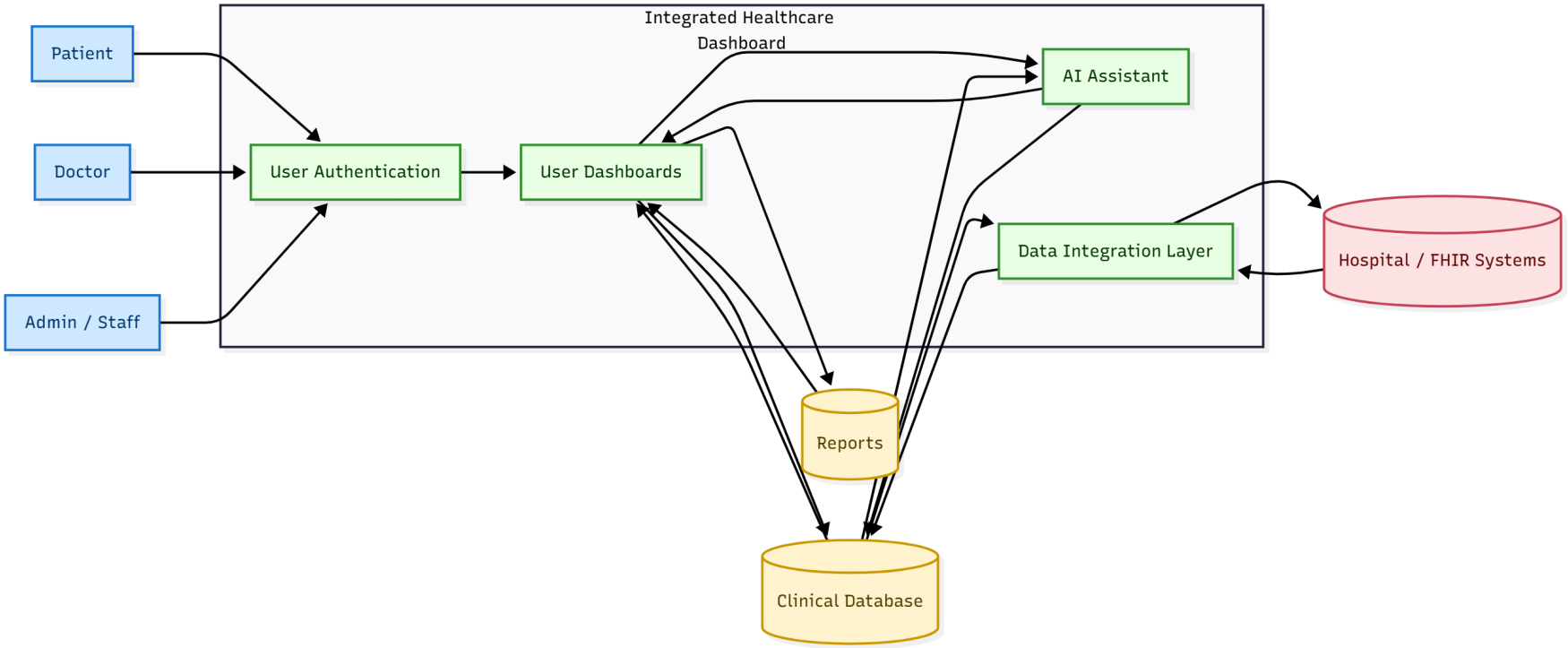
Design Strategy

System Design



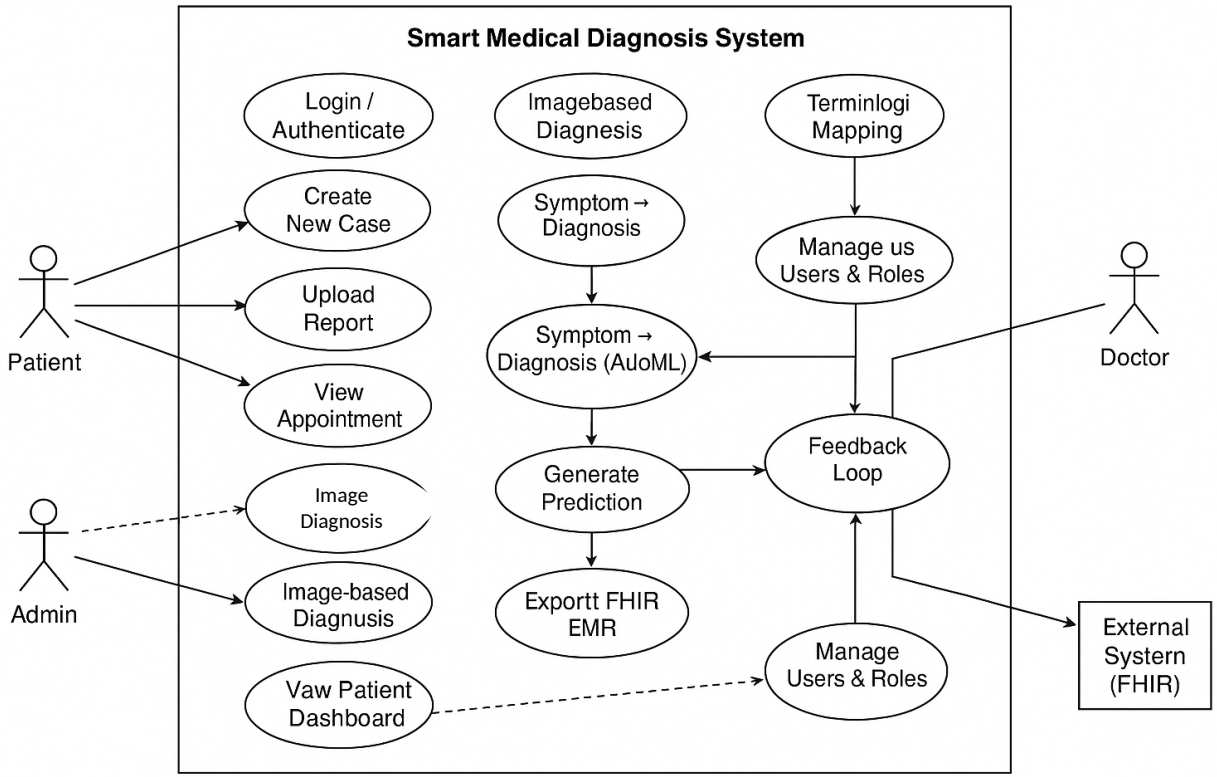
Design Strategy

Data Flow Diagram



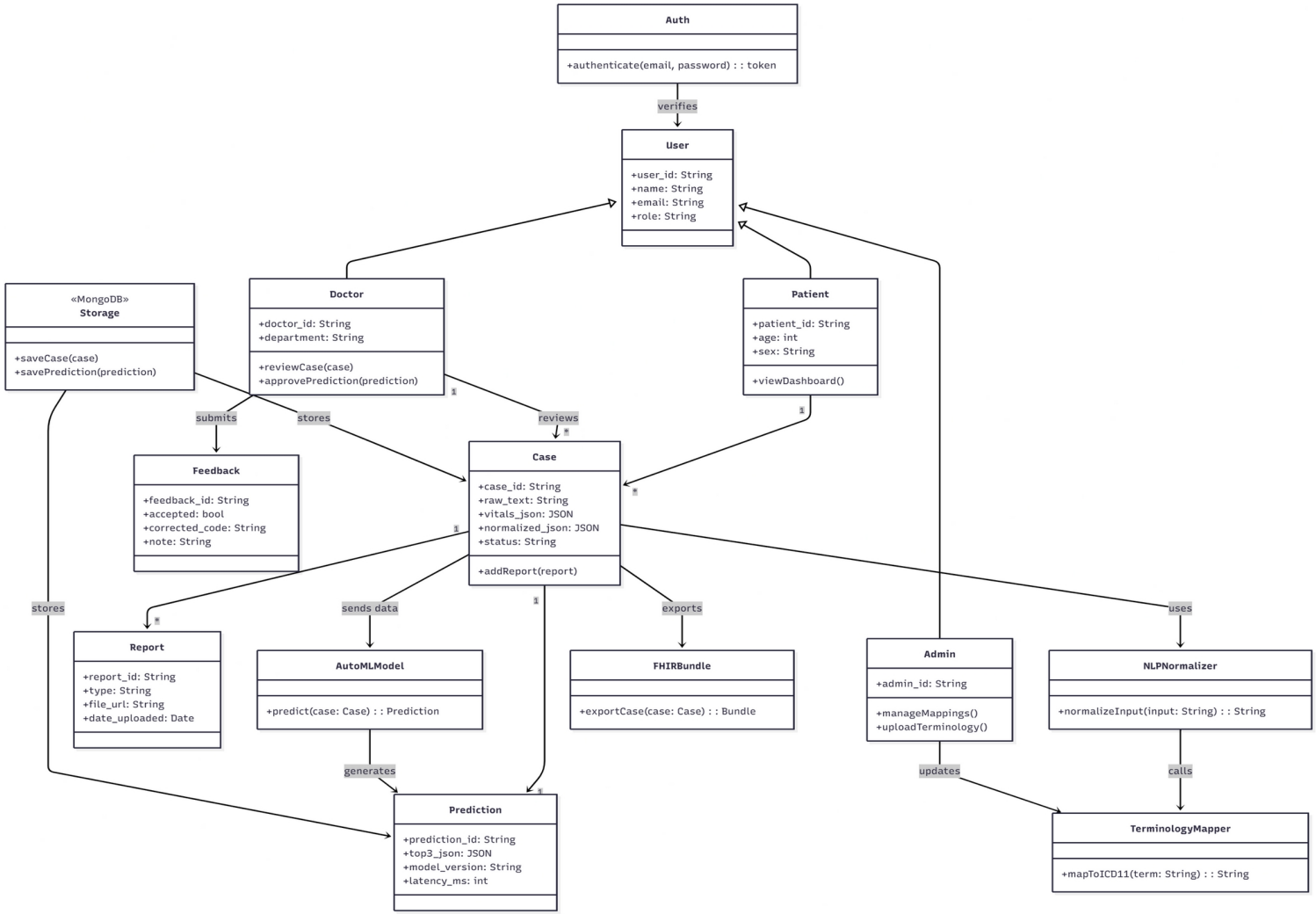
Design Strategy

Use Case Diagram



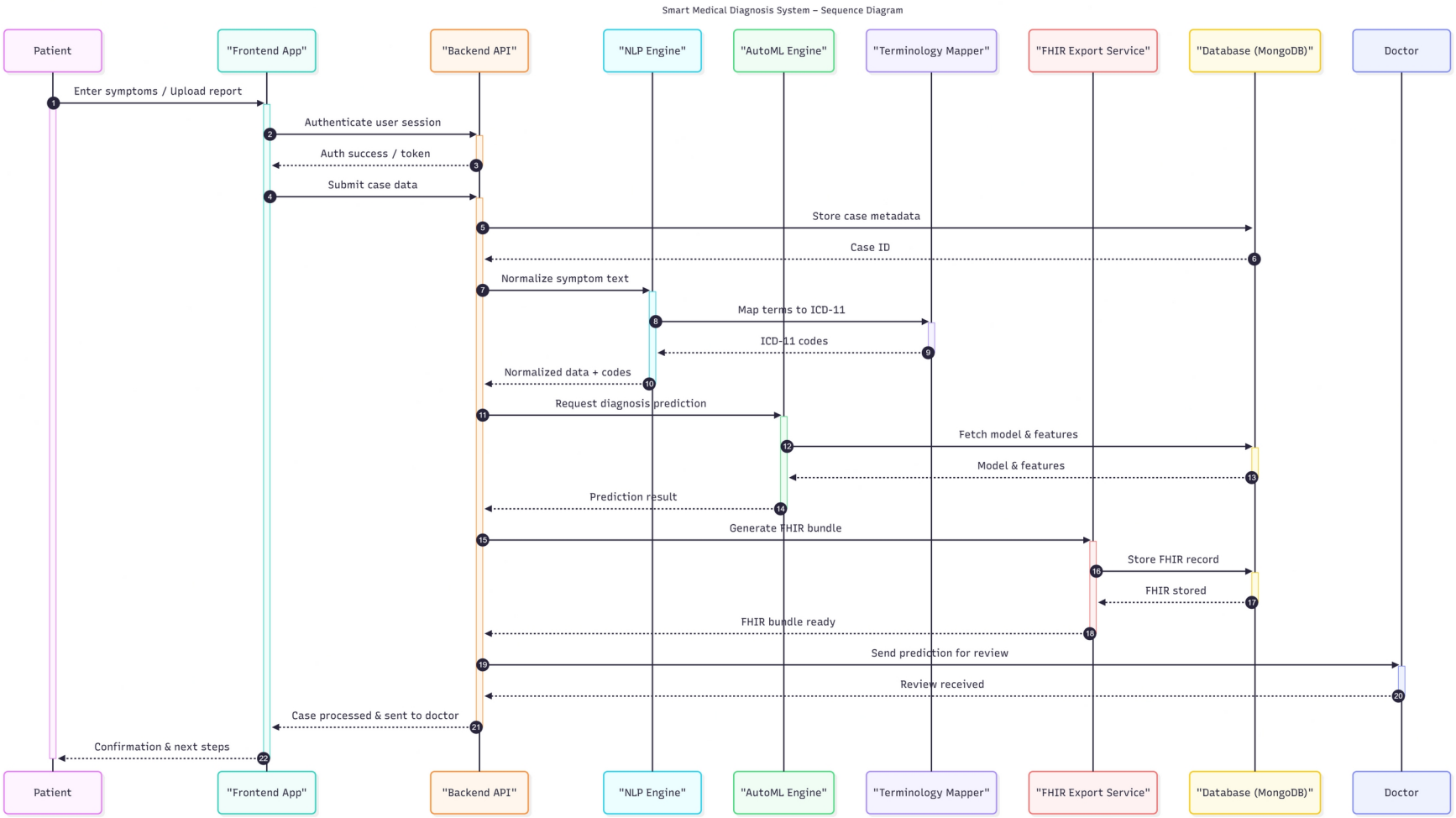
Design Strategy

Class Diagram



Design Strategy

Sequence Diagram



Conclusion

The project has completed its foundational structure, including the core architecture, system workflow design, and UML modelling. These components provide a clear blueprint for development and ensure that the upcoming implementation phases will follow a well-defined technical pathway. With the architecture and design strategy now established, the team is confident about progressing into module development and integration. Once completed, the system is expected to streamline communication between patients and doctors, centralize medical information, and ultimately enhance the overall efficiency and experience of hospital-based patient care.

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